Discussion:
(Un-)orthodoxy in use of HOOK features
Semantic algebra compliance

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MRS Algebra
Copestake et al 2001

- Each semantic composition operator (one per rule) identifies the hook of the argument with some hole in the functor

Definition 6 The definition of op_{subj}

$op_{subj}(a_1, a_2)$ is the following: If $a_1 = \perp$ or $a_2 = \perp$ or $\text{hole}_{subj}(a_2) = \emptyset$, then $op_{subj}(a_1, a_2) = \perp$. And if $\exists l \neq subj$ such that:

$$|\text{hole}_l(a_1) \cup hole_l(a_2)| > 1$$

then $op_{subj}(a_1, a_2) = \perp$. Otherwise:

1. $hook(op_{subj}(a_1, a_2)) = hook(a_2)$
2. *For all labels $l \neq subj$:*
   $hole_l(op_{subj}(a_1, a_2)) = hole_l(a_1) \cup hole_l(a_2)$
3. $lzt(op_{subj}(a_1, a_2)) = lzt(a_1) \oplus lzt(a_2)$
4. $eq(op_{subj}(a_1, a_2)) = Tr(eq(a_1) \cup eq(a_2) \cup \{hook(a_1) = hole_{subj}(a_2)\})$

   where Tr stands for transitive closure.

- For example:

- And that's all we do with the argument's HOOK

Elaborations from Copestake 2007

- For passives:

1. For all labels $l \neq comp1$, $l \neq subj$:
   $slot_l(op_{\text{comp1-subj}}(a)) = slot_l(a)$
2. $slot_{\text{subj}}(op_{\text{comp1-subj}}(a)) = slot_{\text{comp1}}(a)$

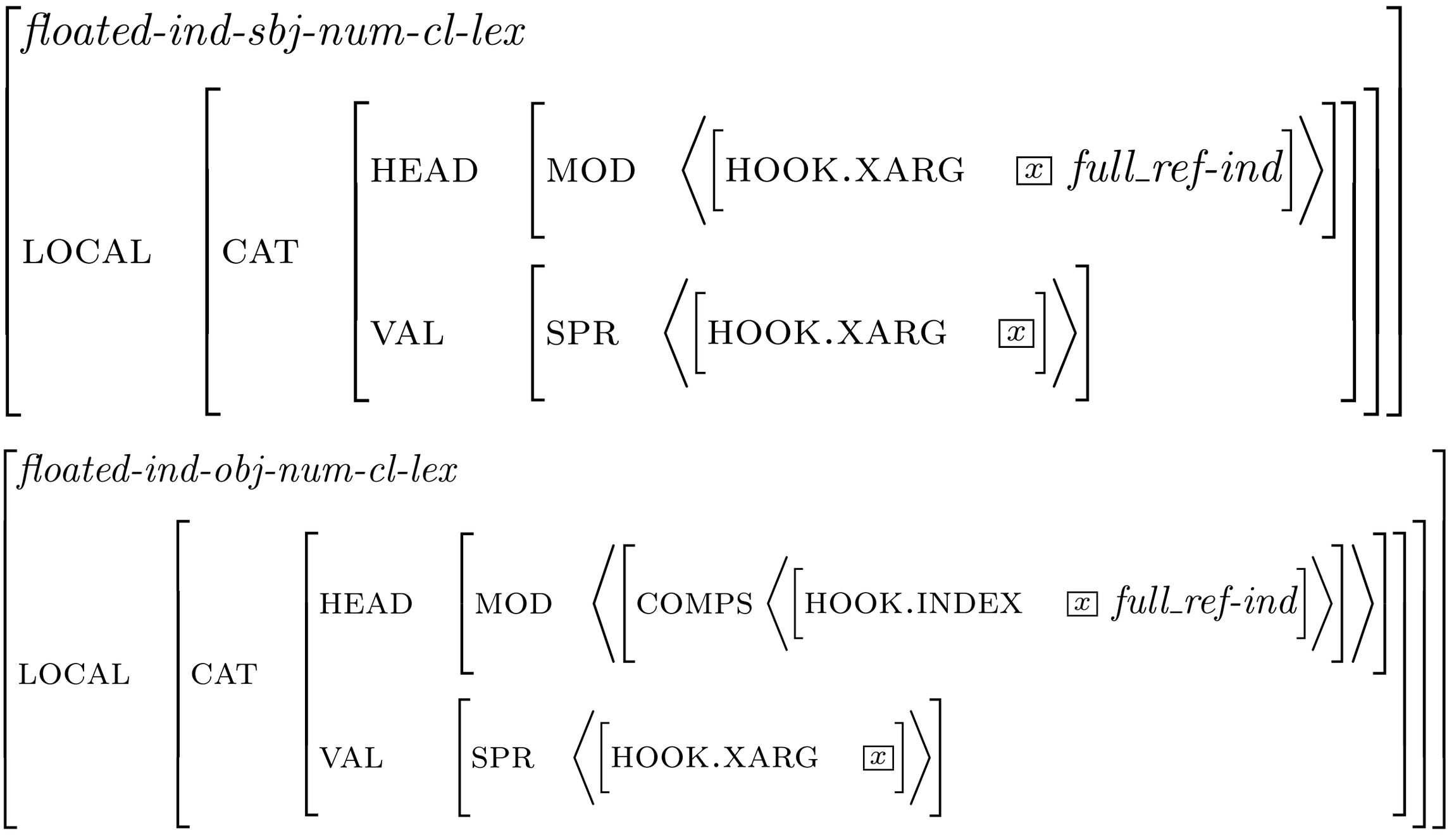
- For optional complements: Construction's sement provides the hook to fill the hole...

Case study #1: Floated numeral classifier phrases in Jacy [jpn] (Siegel et al, forthcoming)

(182) 友達　　が　3　人　　待っています。
tomodachi ga san nin matte imasu
friends NOM three NUMCLF:persons wait progressive
"Three friends are waiting for me."

(183) 友達　　を　3　人　　待っています。
tomodachi wo san nin matte imasu
friends ACC three NUMCLF:persons wait progressive
"I am waiting for three friends."

Case study #1:
Floated numeral classifier phrases in Jacy



Case study #2: Discontinuous NPs in Wambaya [wmb] (Bender 2008)

Babagayi	nyin	jundurra	mirnda
sister.II-LOC	2.SG.A(PR)-PROG	dust.IV(ACC)	1.DU.INC.OBL
bajbaga	yardi.		
big.IV(ACC)	put		

‘Sister you’re making lots of dust for us.’

Case study #2:
Discontinuous NPs in Wambaya [wmb]

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head-2nd-comp-mod-phrase:

$$\left[\begin{array}{l}
\textsc{synsem} \mid \textsc{comps} \left\langle \boxed{1}, \left[\begin{array}{ll}\textsc{opt} & + \\ \textsc{inst} & \boxed{5} \\ \textsc{local} & \boxed{3} \\ \textsc{non-loc} & \boxed{4}\end{array}\right] \right\rangle \oplus \boxed{A} \\
\\
\textsc{hd-dtr} \mid \textsc{comps} \left\langle \boxed{1}, \boxed{2}\left[\begin{array}{ll}\textsc{inst} & \boxed{5} \\ \textsc{local} & \boxed{3} \\ \textsc{non-loc} & \boxed{4}\end{array}\right] \right\rangle \oplus \boxed{A} \\
\\
\textsc{non-hd-dtr} \mid \textsc{mod} \left\langle \left[\textsc{local} \quad \boxed{3}\right] \right\rangle
\end{array}\right]$$

Case study #3: Do-be construction in English (Flickinger & Wasow 2013)

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(1) a. what you have to do is get ready
b. all the government does is send out checks
c. the thing I'm doing is trying to learn from my mistakes
d. the best one can do is compare one risk to the next
e. the least we should do is make it as much fun as possible

E. The copula is followed by a verb phrase.
F. The verb heading the post-copula VP is interpreted as having the same subject as the occurrence of *do* mentioned in C above.
G. The form of the verb heading the post-copula VP depends on

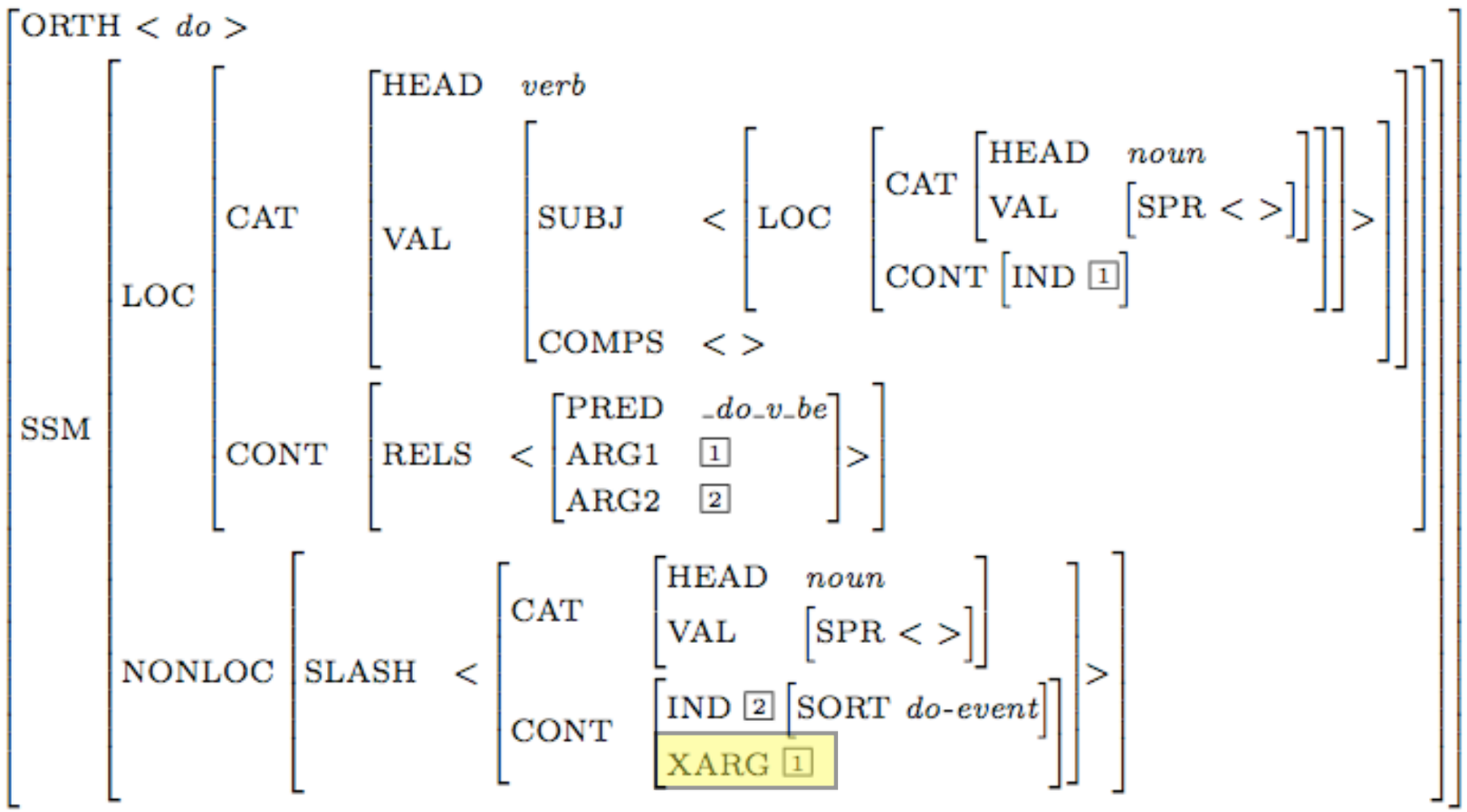


FIGURE 1 Feature structure for DBC verb *do*

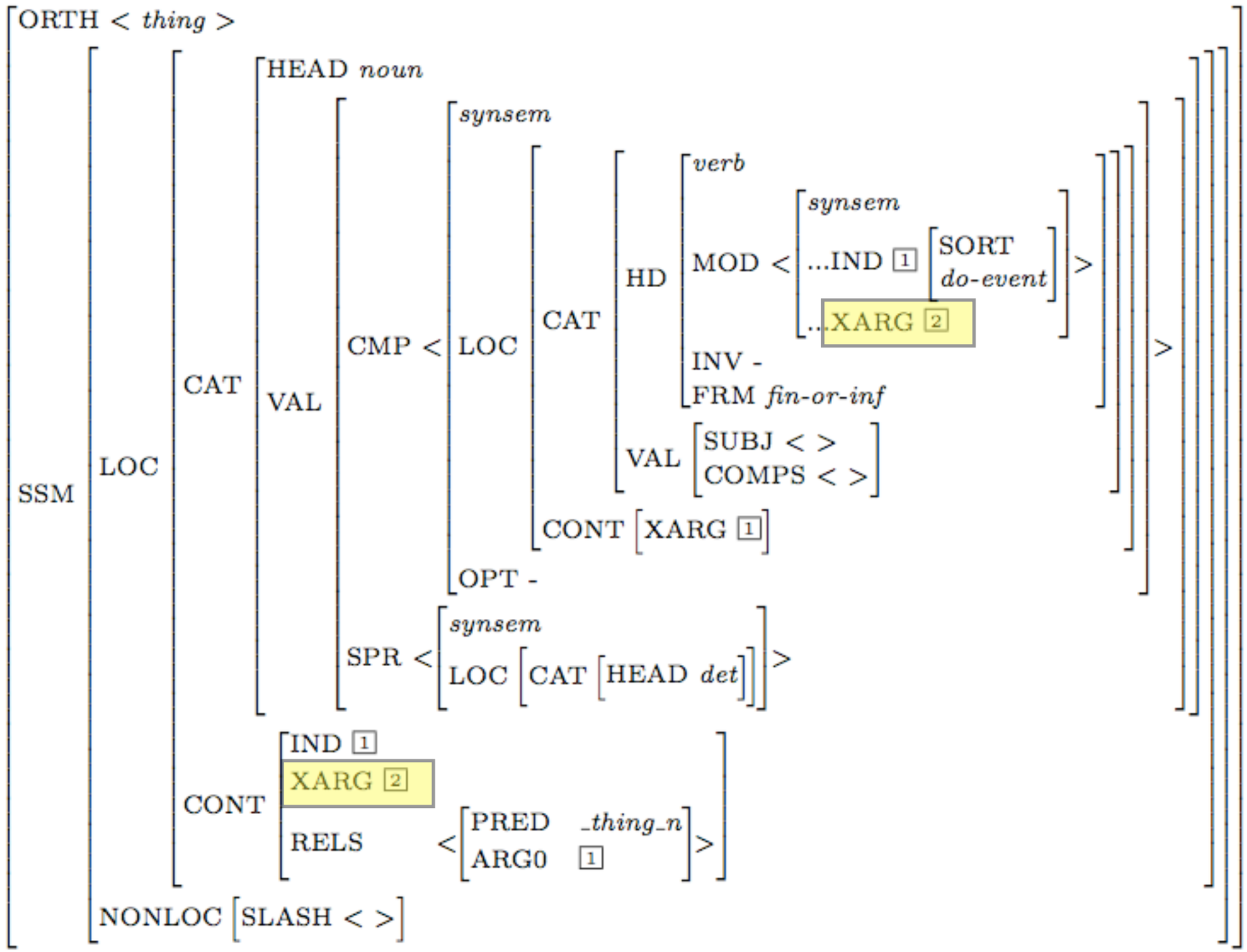


FIGURE 2 Feature structure for DBC noun *thing*

Questions

- Are these analyses in fact consistent with the algebra as developed by Copestake et al 2001?

- If not, should we:
    - Elaborate the algebra to bring them into the fold? (How?)
    - Come up with new analyses?

- More generally: How could one go about testing a grammar for algebra-compliance?
    - Is static analysis possible?